

USHAN PERERA

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Location: Rathnapura, Sri Lanka

PROFESSIONAL SUMMARY

BSc (Hons) in Electronics and Computer Science undergraduate with hands-on experience building real-time secure networks and AI-integrated platforms. Seeking a technical internship to utilize expertise in Software Engineering, API Development, AI & ML and DevOps practices within a fast-paced technology environment.

EDUCATION

BSc (Hons) in Electronics and Computer Science	Jun 2023 - Present
University of Kelaniya Expected Graduation: 2027	

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java, C, C++, C#, PHP
Frontend Technologies: React.js, Next.js, Streamlit, HTML5, CSS3, Tailwind CSS, Bootstrap
Backend Technologies & Frameworks: Node.js, Express.js, FastAPI, Spring Boot, .NET, REST APIs, Pydantic
Databases: MongoDB, MySQL, Oracle, SQL Server, ChromaDB
Cloud, DevOps & Infrastructure: AWS, Microsoft Azure, Docker, Kubernetes, Linux, CI/CD, Prometheus, Grafana
AI & Machine Learning: Pandas, NumPy, LangChain, TensorFlow Lite, Scikit-Learn, Joblib, NLP
Developer Tools & IDEs: Git, GitHub, VS Code, IntelliJ IDEA, Antigravity, Android Studio, Postman, XAMPP

PROJECTS

Sentinel Stream - Real-Time Intrusion Detection System	2026
Tech: React.js, Node.js, Express.js, MongoDB Atlas, Socket.IO, Docker	
- Designed a full-stack intrusion detection system with 4-layer detection mechanisms (volumetric, pattern-based, Z-score, and EWMA) to identify anomalous network behavior in real time. - Built an adaptive feedback system enabling analysts to adjust detection sensitivity through alert classification without requiring model retraining. - Secured the application with JWT authentication, HTTP-only cookies, and Role-Based Access Control (RBAC); deployed the full infrastructure using Docker containerization.	
AI-Powered Smart Inventory System	2026
Tech: React.js, Node.js, Express.js, MongoDB Atlas, Docker, Redis, Prometheus, Grafana, Google Gemini AI	
- Integrated Google Gemini AI into a scalable inventory platform to automate product identification, categorization, and predictive sales forecasting, reducing manual data entry by over 50%. - Implemented monitoring and observability using Prometheus and Grafana to track system performance and business metrics. - Improved backend data retrieval speeds by ~30% through Redis caching and MongoDB aggregation pipelines; automated test workflows via GitHub Actions CI/CD.	

Smart IT Service Management Agent

2026

Tech: FastAPI, Python, LightGBM, Scikit-Learn, LangChain, Google Gemini AI, ChromaDB, Streamlit, Docker

- Engineered a high-performance IT helpdesk agent using FastAPI and Pydantic, integrating a custom LightGBM and Scikit-Learn machine learning pipeline to classify support tickets with ~86% classification accuracy.
- Architected a Retrieval-Augmented Generation (RAG) workflow via LangChain and ChromaDB, utilizing Google Gemini AI to synthesize context-grounded resolution guides from embedded troubleshooting runbooks.
- Implemented a dynamic confidence guardrail (60% threshold) to intelligently route ambiguous requests to Level 2 support, and deployed the full interactive Streamlit interface using Docker containerization.

Travel Commerce Platform | Group Project

2025

Tech: React.js, Spring Boot, MongoDB, REST APIs

- Developed a full-stack digital marketplace connecting travelers with local service providers via a responsive React.js frontend.
- Built secure RESTful APIs with Spring Boot and optimized MongoDB queries, improving dynamic data retrieval efficiency.

EDU QUIZ - Quiz Management System | Group Project

2025

Tech: C#, .NET Framework, Windows Forms, SQL Server

- Developed a desktop application supporting automated quiz evaluation and real-time result generation.
- Architected a centralized SQL Server database with role-based access control to manage multi-tier permissions for 2 user roles (teachers and students).

CERTIFICATIONS

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|---|----------|
| • AWS Cloud Practitioner Essentials - Amazon Web Services | Apr 2026 |
| • Kubernetes for the Absolute Beginners with Hands-on Labs - KodeKloud | Mar 2026 |
| • Docker Training Course for the Absolute Beginner - KodeKloud | Mar 2026 |
| • Hands-on Introduction to Linux Commands and Shell Scripting - IBM | Mar 2026 |
| • Developing Back-End Apps with Node.js and Express - IBM | Mar 2026 |
| • Introduction to Information Security - HashX | Mar 2026 |

EXTRACURRICULAR ACTIVITIES

Membership Coordinator - UOK Robot Battles

Mar 2025 - Aug 2025

- Communicated competition details and updates to registered members through emails and phone calls.
- Provided support to fellow membership coordinators in managing participant engagement activities.
- Assisted the organizing committee in ensuring effective coordination of membership-related operations.

Member - Electronics and Computer Science Club (ECSC)

Jun 2023 - Present

- Contributed to organizing and executing major technical events, boosting student engagement in modern computing.